PRINTER HUB 1.27.0

Setup guide

App, relay, home bridge (NAS · Raspberry Pi · PC) and OrcaSlicer

This takes about 15 minutes. Only part 1 is required. Parts 2–6 add access from away, alerts, OrcaSlicer, slicing from the app and failure detection.

PART	WHAT YOU GET	YOU NEED
1. The app	Control and camera at home	iPhone/iPad or a Mac
2. Relay	The cloud mailbox used for access from away	A free Cloudflare account
3. Home bridge	Access from anywhere, alerts with the app closed	Any always-on box with Docker: NAS, Raspberry Pi (64-bit), PC
4. OrcaSlicer	Send prints from Orca, plus a printer panel inside Orca	Mac app + OrcaSlicer 2.5+
5. Slicing	Slice STL / 3MF / STEP models from your phone, no desktop slicer	The home bridge + ~1 GB of disk
6. Failure detection	An alert (and optionally a pause) when a print turns to spaghetti	The home bridge with a camera + ~600 MB

```
Phone ⇄ relay (Cloudflare) ⇄ home bridge (NAS / Pi / PC) ⇄ printer
         sealed frames only    end-to-end encrypted with your pairing code
```

1. The app

Download from the latest release.

iPhone / iPad: `PrinterHub-x.y.z.ipa`. The file is unsigned, so install it with **SideStore**, **LiveContainer** or AltStore; they sign it with your own Apple ID. Later updates come through the app (Settings → Updates → Update), which hands the new version to LiveContainer or SideStore.

Mac (Apple Silicon): `PrinterHub-x.y.z.dmg`. Open it and drag Printer Hub to Applications. The first time, right-click the app and choose Open. From 1.18 on, the Mac app installs its own updates and asks you to restart.

Add a printer. Tap **+**. Printers on your network are found automatically; otherwise enter the printer's IP address (from the printer's network screen). That's all you need at home.

Supported: anything running Klipper with Moonraker, including Creality K1/K2 (rooted or stock with Moonraker), Snapmaker U1 and Voron.

2. Relay (once)

The relay is a tiny Cloudflare Worker that passes sealed messages between your phone and your bridge. It cannot read them. The free plan is plenty.

1. Install Node.js (LTS) on any computer.
2. Get the relay: download `printer-hub-relay-x.y.z.tar.gz` from the release and unpack it, or use the `relay/` folder of this repo.
3. In that folder:

```
npm install
npx wrangler login      # opens the browser to sign in to Cloudflare
npx wrangler deploy
```

4. Note the address it prints: `https://printer-hub-relay.<you>.workers.dev`.

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3. Home bridge

The bridge runs on a device that stays on and is on the same network as the printer. It keeps one outgoing connection to your relay, so you don't open any router ports. It also sends **alerts** (finished, failed, paused, CFS humidity, low spool) to the free **ntfy** app, even when Printer Hub is closed.

The image is published for **amd64** (most NAS units and PCs) and **arm64** (Raspberry Pi 4/5 with a 64-bit OS, ARM NAS units).

Linux, Raspberry Pi or NAS with SSH: one command

```
curl -fsSL https://raw.githubusercontent.com/dragonemprorx90/printer-hub/main/deploy/install.sh | sh
```

It asks for the printer's IP, your relay URL, whether the printer is a Creality K1/K2 (its camera needs a helper to work from away), and whether to turn on slicing. It then starts everything in `~/printer-hub-bridge` and prints your **pairing code**. Run it again to change answers; your pairing is kept.

Don't have Docker yet? On a Pi or Linux PC: `curl -fsSL https://get.docker.com | sh`, then log out and back in.

NAS with a web UI (UGREEN, Synology, QNAP, Unraid, TrueNAS)

1. Make a folder, e.g. `docker/printer-hub-bridge`.
2. Put the two files from `deploy/` in it: `docker-compose.yml`, and `.env.example` renamed to `.env`.
3. Edit `.env`: set `PRINTER_HOST` and `RELAY_URL`. For a Creality K1/K2, uncomment the two camera lines.
4. In the NAS's Docker app, create a **Project / Stack** from that folder and start it. Synology calls it Container Manager → Project; UGREEN calls it Docker → Project.
5. Open the bridge container's log. The pairing code (`PHB1:…`) and a QR code are printed there.

Windows or Mac PC with Docker Desktop

1. Install Docker Desktop. In Settings → Resources → Network, turn on **Enable host networking**.
2. Download `docker-compose.yml` and `.env.example` from `deploy/` into a folder. Rename the second file to `.env` and edit it as above.
3. In that folder, run `docker compose up -d`, then `docker compose run --rm bridge --show-code`.

A PC only works while it's on and awake. A NAS or Pi is the better home for the bridge.

Without Docker

Each release has plain binaries (`printer-hub-bridge-x.y.z-linux-x64` , `-linux-arm64` , `-macos-arm64`):

```
chmod +x printer-hub-bridge-*
./printer-hub-bridge-* --relay https://printer-hub-relay.<you>.workers.dev --printer 192.168.1.50
```

Pair the app

In the app, open **Edit printer → Remote access code**. Paste the `PHB1:…` code, or scan the QR code. Treat the code like a password. To lock out every phone and issue a new code, run:

```
docker compose down && docker compose run --rm bridge --reset --show-code && docker compose up -d
```

Choose how the app connects

Edit printer → Connection picks the way in, per printer:

CHOICE	WHAT IT DOES
Direct first, then remote access (default)	Talks to the printer's own address when it answers (at home, or over a Tailscale route to your home network). Otherwise it uses the relay, and checks once a minute whether the direct way works again.
Direct only	Never uses the relay: home network or Tailscale only.
Remote access only	Always uses the relay, even at home.

Some networks block the relay (some school, office and hotel Wi-Fi block `workers.dev`). If you use Tailscale, share your home network through it instead:

1. On the machine that runs the bridge: `sudo tailscale set --advertise-routes=192.168.68.0/22` . Use your own home network's range.
2. In the Tailscale admin console, open that machine → **Edit route settings** and approve the route.
3. On your phone or Mac: turn on **Use Tailscale subnets** (CLI: `tailscale set --accept-routes`).

The printer's own address then works from anywhere Tailscale does.

At home the app talks to the printer directly. Away, it goes through the bridge, and it switches back by itself when you return.

More than one printer

Run one bridge per printer: copy the folder, then change `PRINTER_HOST` in the copy's `.env` . Each copy has its own `data/` and its own pairing code.

Alerts

On first start the bridge log shows an **ntfy topic**. Install ntfy (iOS, Android) and subscribe to that topic. To set quiet hours, humidity and spool limits, or turn alerts off, see the options in `.env` .

Updating the bridge

```
cd ~/printer-hub-bridge && docker compose pull && docker compose up -d
```

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4. OrcaSlicer (Mac)

Printer Hub on the Mac runs a small local service on `127.0.0.1:7126` (Settings → Orca Slicer).

Send prints. In Orca, open the printer's connection settings:

- Host type: **Octo/Klipper**
- Hostname: `127.0.0.1:7126`. To skip the printer question, add the printer's name, e.g. `127.0.0.1:7126/k2-plus`.

Press **Print** or **Send**. Printer Hub uploads the file (at home or through the bridge) and opens print setup.

Printer panel (Orca 2.5+). Download `printer_hub.py` from the release, or save it from Settings → Orca Slicer. Install it from Orca's **Plugins** dialog. A panel opens beside the 3D view with every printer: camera, progress, temperatures, controls, filaments, spools, history and rules.

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5. Slicing from the app

Pick an **STL, 3MF, STEP or OBJ** model in **Files → Upload** (or drop it on the Mac window). First choose where to slice it (below). **Slice in the app** opens the slice editor with the model on your printer's plate. It works at home and away.

Where to slice. A model from Find models or Files → Upload asks where it goes. The app remembers your last choice on each device and lists it first; until you pick another, it's **Slice in the app**.

- **Slice in the app:** the slice editor below, on your home bridge.
- **Open in OrcaSlicer** (Mac): the model is saved to `~/Downloads/Printer Hub` and opens in OrcaSlicer, whether Orca is running or not. Shown when OrcaSlicer is in Applications.
- **Send to my Mac's OrcaSlicer** (iPhone): the model goes to a hidden `.printerhub-inbox` folder on the printer, at home or through the encrypted relay. The Mac app checks every printer's inbox every 20 seconds, opens what it finds in OrcaSlicer and deletes it from the printer. The Mac app must be running with OrcaSlicer installed; until then the file waits on the printer. The home bridge needs no update for this.

- **See it:** drag to turn the view, pinch or scroll to zoom, two fingers to pan. The model turns red when it doesn't fit the build volume.
- **Place it:** drag the model to move it. Turn it about X, Y or Z (15° or 90° steps), scale it, use **Lay on face** and tap the face that should rest on the plate, or use **Auto-orient** (Orca picks the orientation). **Centre** puts it back in the middle. The model is sliced exactly where you put it.
- **Print settings:** start from an Orca quality preset and change only what you need: layer height, walls, infill and its pattern, supports (normal or tree, overhang angle, from the plate only), brim, wall and infill speeds, flow ratio, and nozzle and bed temperature (for the toolhead that prints, on the U1). Empty fields keep the preset. The app remembers your changes per printer; **Use preset** clears them.

- **Slice:** you see the print time, the filament in grams and every layer. **Continue to print** opens the file with its **Print** button.

A 3MF you don't move is sliced as it is, with its painted colours and plates. Once you move it, the mesh as placed is sliced and the colours are not kept. STEP, 3MF and OBJ are turned into a mesh by the slicer container for the 3D view; STL is read on the phone.

```
App —(encrypted relay)→ home bridge → slicer container (OrcaSlicer 2.5) → G-code to the
printer (LAN)
```

Slicing runs in a second container next to the bridge, using OrcaSlicer's own Creality K2 Plus and Snapmaker U1 profiles (0.4 mm nozzle). On the U1 you pick a filament for each of the 4 toolheads and which one prints the model. A 3MF with painted colours keeps them. Your model never leaves your home except inside the encrypted relay, and nothing is uploaded to a slicing service.

Turn it on. Re-run the installer and answer **y** to "Slice models from the app?". Or add `slicer` to `COMPOSE_PROFILES` in `.env` (e.g. `COMPOSE_PROFILES=slicer`, or `creality,slicer` for a K2), then:

```
docker compose pull && docker compose up -d
```

- The image is about 1 GB, for amd64 and arm64. A slice takes 10 s to a few minutes; a NAS or Pi 5 is fine, a Pi 4 is slow.
- **Two printers on one machine:** each bridge folder runs its own slicer. Give the second one `SLICER_PORT=7071` in its `.env`.
- **Slice without the relay:** the slicer only listens on `127.0.0.1` (for the bridge). To reach it straight from the app, for example over Tailscale, add the machine's Tailscale address in `.env`: `SLICER_BIND=127.0.0.1,100.x.y.z`. Then restart with `docker compose up -d`, and in the app set **Edit printer → Slicer address** to `100.x.y.z:7070` (or `:7071`). The app uses that address when it answers and falls back to the relay, following the printer's **Connection** choice. Anyone who can reach that address can slice and send G-code to the printer, so don't use `0.0.0.0` or a public address.
- **Your own presets:** in desktop Orca, export a process or filament preset and copy the `.json` into `slicer-profiles/process/` or `slicer-profiles/filament/` in the bridge folder. They show up in the app's lists.
- **Find models online:** in Files, the globe button searches **Printables** (no account), **Creality Cloud** (search is open; downloads ask you to sign in once on Creality's own page inside the app) and **Thingiverse** (paste a free app token from thingiverse.com/developers). **MakerWorld** opens as its own website inside the app: sign in there, tap Download on a 3MF, and Printer Hub slices it. G-code or a 3MF already made for your K2 Plus or U1 is sent straight to the printer.
- STEP files are meshed first (fine detail is kept to about 1/36 of a circle), which adds up to a minute for big parts.

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6. Failure detection

The home bridge can watch the camera during a print and warn you when it looks like spaghetti: a push alert with the picture (with the app closed too), and an alert in the app. It can also pause the print.

```
camera → home bridge (a frame every 30 s while printing) → detector container → score
                    └→ alert (ntfy + app), pause through Moonraker if you turned it on
```

The detector is Obico's open failure model (The Spaghetti Detective), run on the CPU. It looks at a frame in under a second on a NAS, so 2 frames a minute is light work, and nothing leaves your home. One frame never alerts on its own: the score must stay high for about a minute.

Turn it on. The bridge needs a camera (`CAMERA_URL`, see part 3). Re-run the installer and answer **y** to "Spot failed prints?". Or add `detector` to `COMPOSE_PROFILES` in `.env` (e.g. `creality,detector` for a K2), then:

```
docker compose pull && docker compose up -d
```

- **Pause or alert only:** alert only by default. While a print runs, the printer's screen in the app shows **Failure watch** with a **Pause if seen** switch for that printer. The bridge pauses through Moonraker; resume or cancel from the app.
- **App alerts:** Settings → Notifications → **Possible print failure**.
- **Two printers on one machine:** one detector serves both bridges. Add `detector` to the profiles in one folder only.
- **False alarms:** raise `FAILURE_THRESHOLD` in `.env` (default 0.8; 1.2 is stricter). It judges only what the camera sees, so aim the camera at the bed.
- The image is about 550 MB, for amd64 and arm64, and uses up to 1 GB of memory.

Troubleshooting

PROBLEM	FIX
Printer not found	Phone and printer must be on the same Wi-Fi. Enter the IP by hand. Moonraker must be on port 7125 (or set the port).
K2 camera shows "sent no video"	The camera serves one viewer at a time. Close the Creality app or Creality Print on other devices, wait 30 s, then tap Retry.
Camera from away is black (K1/K2)	`.env` needs both `COMPOSE_PROFILES=creality` and the `CAMERA_URL` line. Check `docker compose logs go2rtc`.
App says the bridge is offline	`docker compose logs bridge`. Check that `RELAY_URL` is your own relay and that the NAS can reach the internet.
"Update the slicer on your home bridge to view this model" or "...to place the model and change print settings"	The slicer container is older than the editor (1.24). Run `docker compose pull && docker compose up -d` in the bridge folder.
"The slicer isn't running on your home bridge"	Update the bridge (`docker compose pull && docker compose up -d`) and add `slicer` to `COMPOSE_PROFILES`. Check `docker compose logs slicer`.
Find models: Creality download asks you to sign in again	Creality ended the session. Sign in again on its page in the app (Settings → Model sites shows the state).
Find models: "Thingiverse refused the token"	Create a new app at thingiverse.com/developers, then use the key button in Find models → Thingiverse to paste its App Token.
MakerWorld: tapping Download does nothing	Sign in to MakerWorld on its page first. Printer Hub catches 3MF, STL and STEP downloads only.
No failure alerts	`docker compose logs bridge` should not say "Failure detector not reachable". Check `docker compose logs detector`, that `detector` is in `COMPOSE_PROFILES` and that `CAMERA_URL` is set.
Lost the pairing code	`docker compose run --rm bridge --show-code`
Mac update didn't install	Printer Hub must live in Applications, and your user must be allowed to change it (admin). Otherwise the app opens the download.

Still stuck? Open an issue with the printer model, app version and the bridge log.